Source-Based AI Recommendation Explanations — A Priori Power Analysis v1

1. Purpose

An a priori power analysis was conducted to estimate the sample size required for the planned five-condition between-subjects experiment.

The primary experimental factor is **Recommendation Explanation Source**, with five levels:

1. No-source Control
2. Behavior-Based
3. Review-Based
4. Product-Feature-Based
5. Mixed-Source

The primary experimental analyses will first evaluate overall differences among the five conditions using one-way ANOVA, followed by theoretically specified planned contrasts.

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2. Primary Power-Analysis Assumptions

The initial sample-size calculation uses the following assumptions:

- Statistical test: One-way ANOVA
- Number of groups: 5
- Significance level: $\alpha = .05$
- Statistical power: $1 - \beta = .80$
- Assumed effect size: Cohen's $f = .20$
- Allocation: Approximately equal across conditions

The assumed effect size of **f = .20** is treated as a design-stage small-to-moderate effect-size assumption rather than an empirical estimate from the present study.

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3. Required Analyzable Sample

Under these assumptions, the estimated minimum total sample size is approximately:

N = 304 participants

To maintain approximately equal allocation across five conditions, the study will use a minimum balanced analyzable target of:

N = 305

or approximately:

61 participants per condition

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4. Recruitment Target

Some observations may need to be excluded because of incomplete responses, failed data-quality checks, duplicate responses, or other prespecified exclusion criteria to be finalized before data collection.

Therefore, the initial recruitment target will be:

N = 360 participants

This corresponds to approximately:

72 participants per condition

This target provides a buffer above the minimum analyzable sample while maintaining balanced allocation across conditions.

Reproducibility. Sample-size calculations will be reproduced in the project repository using Python power-analysis code.

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5. Effect-Size Sensitivity

The required sample size depends substantially on the true magnitude of the experimental effect.

Approximate total sample requirements for a five-group one-way ANOVA with $\alpha = .05$ and power = .80 are:

- $f = .15 \rightarrow$ approximately N = 536
- $f = .20 \rightarrow$ approximately N = 304
- $f = .25 \rightarrow$ approximately N = 196
- $f = .30 \rightarrow$ approximately N = 138

Thus, the planned sample of approximately 360 participants should not be interpreted as providing adequate power for arbitrarily small effects.

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6. Planned-Contrast Consideration

Several hypotheses involve specific comparisons between two experimental conditions rather than only the five-group omnibus effect.

With approximately 72 participants per condition, a two-group comparison has approximately 80% power to detect a standardized mean difference of roughly:

Cohen's d ≈ 0.47

Smaller pairwise effects would require a larger sample.

For example, detecting an effect around **d = 0.40** with approximately 80% power would require roughly 100 participants per condition, corresponding to approximately:

N = 500 total participants

Therefore, the final main-study sample may be increased if prior empirical evidence, pilot results, or resource availability suggests that smaller effects should be expected.

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7. Mediation and Indirect Effects

The present power calculation is based primarily on the experimental condition effect and does not establish that the study is adequately powered for every proposed indirect effect.

The proposed mediation pathways include:

- Behavior-Based → Perceived Relevance → Trust → Reliance
- Behavior-Based → Privacy Concern → Trust → Reliance
- Review-Based → Source Credibility → Trust → Reliance
- Product-Feature-Based → Perceived Relevance → Trust → Reliance

Indirect effects will be evaluated using bootstrap confidence intervals.

A more specific power analysis for mediation may be conducted after pilot information or defensible estimates of the component path coefficients become available.

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8. Current Planning Decision

Minimum analyzable sample

N = 305

Initial recruitment target

N = 360

Conditions

5

Approximate recruitment per condition

72 participants

Primary design-stage assumption

Cohen's f = .20, α = .05, power = .80

The power analysis will be revisited before preregistration of the final main study if pilot evidence or prior literature provides a stronger basis for selecting an expected effect size.